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ECE 662

Project 2

To implement the UNet classifier, I took the UNet developed in class and extended it with a pooling step followed by an MLP head to generate the final logits. I chose to use an optax adamw solver with weight decay and a cross entropy loss function.

I found that the initial implementation (run_CIFAR_UNet.py) worked, but quickly overfit the train dataset and only reached a test accuracy of ~63% despite a very high train accuracy. In order to help mitigate the overfitting I added in data augmentation for the train dataset (run_CIFAR10_UNet_aug.py). Each epoch, a new train dataset is used with randomized augmentations (RandomCrop and RandomHorizontalFlip). This substantially improved the model's ability to generalize and performance on the test dataset. After 100 epochs, the model reached an accuracy of ~84% on test data.

I was curious to see which classes the model was confused on, so after training I saved the model and generated a heatmap confusion matrix using the CIFAR-10 test dataset. Most of the confusion in the model made sense, for example dogs and cats were often confused with one another

Model performance after 30 epochs (overfitting) without augmentation

```
...
Epoch 20, Loss: 0.4557, Train Acc: 84.22%, Test Acc: 60.79%
Epoch 21, Loss: 0.3508, Train Acc: 87.10%, Test Acc: 61.44%
Epoch 22, Loss: 0.0892, Train Acc: 90.21%, Test Acc: 62.32%
Epoch 23, Loss: 0.0246, Train Acc: 89.41%, Test Acc: 61.94%
Epoch 24, Loss: 0.0254, Train Acc: 90.42%, Test Acc: 61.71%
Epoch 25, Loss: 0.0225, Train Acc: 91.71%, Test Acc: 61.90%
Epoch 26, Loss: 0.0185, Train Acc: 91.97%, Test Acc: 62.26%
Epoch 27, Loss: 0.5734, Train Acc: 91.25%, Test Acc: 62.16%
Epoch 28, Loss: 0.1915, Train Acc: 90.33%, Test Acc: 61.89%
Epoch 29, Loss: 0.0157, Train Acc: 92.32%, Test Acc: 61.62%
Epoch 30, Loss: 0.0030, Train Acc: 93.67%, Test Acc: 63.69%
```

Model performance after 100 epochs with data augmentation

```
...
Epoch 90, Loss: 0.2816, Train Acc: 90.66%, Test Acc: 83.26%
Epoch 91, Loss: 0.2750, Train Acc: 90.27%, Test Acc: 82.55%
Epoch 92, Loss: 0.2753, Train Acc: 90.94%, Test Acc: 84.34%
Epoch 93, Loss: 0.2756, Train Acc: 90.82%, Test Acc: 83.88%
Epoch 94, Loss: 0.2745, Train Acc: 90.76%, Test Acc: 83.48%
```

```
Epoch 95, Loss: 0.2678, Train Acc: 91.00%, Test Acc: 83.15%
Epoch 96, Loss: 0.2695, Train Acc: 91.22%, Test Acc: 84.38%
Epoch 97, Loss: 0.2646, Train Acc: 90.12%, Test Acc: 83.23%
Epoch 98, Loss: 0.2661, Train Acc: 89.62%, Test Acc: 81.58%
Epoch 99, Loss: 0.2647, Train Acc: 90.92%, Test Acc: 83.55%
Epoch 100, Loss: 0.2580, Train Acc: 91.72%, Test Acc: 83.96%
Best test accuracy: 84.38%
```